

Def. A continuous real-valued $f: D \rightarrow \mathbb{R}$ on a domain $D \subseteq \mathbb{C}$ is called harmonic if:
 f is partially twice continuously differentiable ($f \in C^2$) satisfying

$$f_{xx} + f_{yy} = 0. \quad (\text{It is called the Laplace's equation.})$$

Thm. Let $f = u + iv$ be an holomorphic on a domain D .

Then, both u and v are harmonic on D .

Proof.